

MICHAEL TUPAS CARDOSE

SOFTWARE ENGINEER / FULLSTACK DEVELOPER / MOBILE APP DEVELOPER

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PROFESSIONAL SUMMARY

Full-stack software engineer who ships production systems end to end — React 18/19 + TypeScript front ends, React Native app published to the App Store, and PHP, Node, and Cloudflare Workers back ends over MySQL and D1. Rebuilt the public flow of a legacy PHP admission system as a standalone React SPA that processed ~5,000+ (and counting) live applicant registrations, closing a SQL-injection surface and an exam-center overbooking race on the way; sole engineer on a kosher restaurant discovery platform spanning a React Native app, a Next.js admin dashboard on Cloudflare Workers, and its public API. Comfortable owning the work most front-end developers hand off: schema and migrations, session auth, concurrency and rate limiting, WCAG accessibility.

TECHNICAL EXPERTISE

- Programming Languages: PHP, JavaScript, C++, COBOL, Visual Basic, Java
- Frontend Development: JavaScript (ES6+), TypeScript, React.js, Next.js, Vue.js, HTML5, CSS3, Tailwind CSS, DOM Manipulation
- Backend & Database: Laravel, Eloquent ORM, RESTful API design, Node.js, Express.js, PHP 8, CodeIgniter, Laravel, PDO, PSR-4, REST APIs, MongoDB, Microsoft SQL, MySQL, MySQL spatial (POINT/SRID 4326, SPATIAL INDEX, ST_Distance_Sphere), database migrations, transactions & row-level locking, Cloudflare Workers, D1 (SQLite), Durable Objects, Drizzle ORM, wrangler migrations.
- Mobile Development: React Native (bare workflow, not Expo), styled-components/native, react-native-maps, AsyncStorage, Reanimated/gesture-driven sheets, OneSignal push, Firebase, CocoaPods/Gradle native builds, Metro.
- UI/UX & Interactive Design: shadcn/ui, Aceternity UI, Framer Motion, mobile-first responsive design, cross-browser compatibility
- Payments & Third-Party APIs: Stripe (Checkout, Payment Intents, webhooks, Connect Express payouts, refunds), Google OAuth 2.0 / Identity Services, Google Calendar API, PHPMailer/SMTP, Leaflet & OpenStreetMap/Nominatim
- Real-Time & Security: WebSockets (Workerman, Socket.io), presence & typing indicators, JWT/HttpOnly cookie sessions, CSRF double-submit, CSP & security headers, rate limiting, idempotency keys, structured logging with correlation IDs, token-bucket rate limiting via Durable Objects, PBKDF2-HMAC-SHA256 session auth over Web Crypto, SSRF-safe URL validation.
- Web Hosting & Deployment: Apache 2.4 (mod_rewrite, ModSecurity, php.ini tuning), Vercel, DigitalOcean (Linux), Nginx, PM2, SSH/Putty, WinSCP
- Tools & Workflow: Git, GitHub, npm, pnpm, Vite, Webpack, add *PHPUnit*, *Vitest*, *GitHub Actions (CI/CD)*, *Role-Based Access Control (RBAC)*, Yarn 1, pnpm workspaces, Vitest (@cloudflare/vitest-pool-workers), Jest, i18n catalog management (12 locales)
- WCAG accessibility: axe-core, ARIA patterns, keyboard navigation, contrast auditing, accessibilityState, Reduce Motion, screen-reader announcement composition.

- SEO & Analytics: Technical SEO (meta tags, structured data, sitemaps), Google Analytics 4, Google Tag Manager, Google Search Console, Bing Webmaster Tools
- Performance Optimization: Core Web Vitals, lazy loading, code splitting, WebP image conversion, Lighthouse & PageSpeed Insights
- CMS & Backend Integration: Sanity.io (headless CMS), WordPress (custom themes/plugins), Elementor, Divi Builder, Astra, Beaver Builder
- Other Skills: Figma, Photoshop, Canva, wireframing, basic UI/UX design

WORK EXPERIENCE

Software Engineer | Maritime Academy of Asia and the Pacific (MAAP) — Information Systems & Services Department

June 2026 – Present

- Legacy system modernization: rebuilt the two public applicant-facing pages of MAAP's legacy PHP admission system (OMAS) as a standalone React 18 + TypeScript SPA
 - Frontend development: built a 7-step application wizard (101 validated fields) with React Hook Form + Zod v4, schema gating, and draft autosave (localStorage for text, IndexedDB for file uploads) with restore-on-return.
 - Backend & API development (PHP): designed a layered JSON API with PSR-4 autoloading, thin endpoint controllers, and Repository/Service classes; replaced legacy string-concatenated SQL with PDO prepared statements to close a SQL-injection surface on the public registration endpoint; fixed an exam-center overbooking race condition.
 - Email & notifications: built a branded HTML confirmation email (PHPMailer over SMTP), delivered as best-effort after transaction commit so SMTP issues can never roll back a saved registration.
 - Deployment & production support: authored the server runbook covering Apache, PHP, MySQL, SMTP, and filesystem permissions.

Mobile App Developer / Full-Stack Developer | Kosher Kav LLC — USA (Remote)

August 2026 – Present (Project-Based) Sole engineer.

- Kosher restaurant discovery, delivery and video platform — a React Native app plus a Cloudflare Workers admin dashboard and public API.
- Vendor-lock breakout: the product shipped as a white-label tenant of ordering.co's marketplace template (ordering.co/kosherkav), whose hosted builder rebuilds from their template. Forked the React Native 0.81 / React 19 codebase into a repo we control, vendored the headless @components layer in so the tree clones and builds standalone, and took full ownership of build, signing and store publishing.
- Architected a dual-backend architecture to support kosher data without modifying ordering.co's rigid schema.
- Admin dashboard & public API (second codebase): Next.js 16 (App Router, Turbopack) + React 19 + Tailwind v4 on Cloudflare Workers via @opennextjs/cloudflare, with Cloudflare D1 and Drizzle ORM migrations, a pnpm workspace (apps/dashboard, packages/shared, packages/db). Tested inside workerd with Vitest + @cloudflare/vitest-pool-workers against real D1.
- Durable Objects for counting and abuse control: one ViewCounter per video buffering plays and flushing to D1 on a 10-second alarm (SQLite-backed).

- Videos tab (new product surface, built end to end): feed with city rails, featured video, kosher Meat / Dairy / Parev filters that gate the whole screen, type-as-you-go search, a full-screen swipe-to-next player, share-to-web deep links, geolocation-based city. Delivered with the client's seven change requests, each verified on a release APK.
- Map discovery experience: data-driven kashrut chip row, multi-select Hashgacha (certifier) sheet, two-column card grid with a Map⇌List toggle, sort dropdown, Directions deep link, route drawing over a hand-written Google encoded-polyline decoder.

WordPress Specialist (Freelance) | eventsbytmss.com — United States (Remote)

May – June 2026 (Contract)

- Performance & SEO optimization: conducted comprehensive site audits to identify bottlenecks; drastically improved page speed and eliminated Cumulative Layout Shift (CLS) using native browser lazy-loading with CSS animation skeletons; integrated Google Analytics (gtag.js) with deferred loading to preserve site performance.
- Advanced frontend development: engineered a custom, dynamic gallery lightbox using vanilla JavaScript (MutationObserver, localStorage) to manipulate the DOM and enable seamless cross-page data transfer, improving the user journey from gallery to contact form.
- Responsive UI/UX overhaul: led the structural redesign of core pages (Home, About, Services, Gallery, Appointments); enforced a strict typography hierarchy, ensured pixel-perfect mobile/tablet responsiveness, and resolved complex CSS stacking conflicts including off-canvas mobile navigation.
- Custom booking system integration: deployed and customized a booking system with custom fields, consent checkboxes, and JS-driven placeholders; used PHP to automate "Add to Calendar" functionality, dynamically injecting attendee emails via URL query parameters.
- Technical troubleshooting & maintenance: managed site backups, resolved broken redirects, debugged plugin conflicts, and applied custom CSS/JS overrides to refine theme components and improve accessibility.

Front-End Developer | Central One Bataan Inc.

April 2024 – August 2024 (Retrenchment)

- Frontend development: built and maintained responsive websites using HTML, CSS, JavaScript, TypeScript, React.js, Next.js, and Vue.js, styled with Tailwind CSS.
- UI & CMS integration: implemented shadcn/ui, Framer Motion, and Aceternity UI components; integrated Sanity headless CMS and managed custom WordPress themes/plugins.
- Performance optimization: improved page load times through code splitting, lazy loading, efficient bundling, and WebP image conversion; ensured cross-device responsiveness.
- SEO & analytics: implemented technical SEO (meta tags, Schema.org structured data, XML sitemaps, robots.txt), managed Google Search Console, and configured GA4 and GTM for user behavior and event tracking.

Front-End Developer (Freelance) | Suman Gupta

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August 2024 – June 2026 (absorbed from previous employer's client)

- Continued full front-end ownership: delivering the same scope of responsive web development, performance optimization, and SEO/analytics work as above on a freelance, project basis.

WordPress Developer (Freelance) | Randall J. Borden Law Firm

August – September 2025 (Project-Based)

- Environment setup & migration: configured a local WordPress environment for safe development, worked within the Theme File Editor, and migrated the live site into WP Local for secure testing.
- Redesign & optimization: led navigation redesign and responsive layout fixes, developed blog content, optimized logo assets, and added gradient/hover styling to the homepage.
- SEO, analytics & tooling: updated meta titles/descriptions, verified sitemap and robots.txt, integrated GTM, Google Analytics, and Bing Webmaster Tools, and built a Daily Time Tracker in Google Sheets with automated chart visualization.

WordPress Developer (Freelance) | HIREAWIZ Web Design & Digital Marketing

October – November 2025 (Project-Based)

- Page building: built and maintained WordPress pages using Beaver Builder with clean layout hierarchy and modern UI design matching brand guidelines.
- Customization: customized modules and global styling, integrated custom CSS for advanced styling beyond Beaver Builder's native capabilities, and fixed responsiveness issues across devices.

EDUCATION

Bachelor of Science in Information Technology

Polytechnic University of the Philippines | 2017 – 2021

Internship

Barangay Alasasin | July – September 2021

TRAININGS & CERTIFICATIONS

Advance React - Coursera | Maritime Academy of Asia and the Pacific

August 2026

Full Stack Training – MERN | ASG Platform

March 2023 – June 2023

- Completed a full web application project using Node.js, Express.js, React.js, and MongoDB, including DigitalOcean server hosting.
- Covered JavaScript, HTML5, CSS3, Git, Node.js security, RESTful APIs, React Hooks, state management with Redux, and Puppeteer.

NOTABLE PROJECTS

- GlowSimcha (B2C Booking Marketplace, Github Repo is Private) — Two-sided booking platform connecting clients with beauty artists (hair, makeup, nails). Framework-free PHP 8.2 API (PSR-4, service-layer architecture, PDO prepared statements; 87 REST endpoints, 40 migrations, 27 tables) with a React 19 + TypeScript (strict) SPA on Vite, Tailwind v4, and shadcn/ui. Payments via Stripe Checkout with signature-verified

webhooks, an idempotency ledger, deposit/full-payment modes, policy-driven refunds, and Stripe Connect Express payouts (destination charges with proportional transfer reversal). Self-hosted WebSocket server (Workerman) delivering live messaging, typing indicators, presence, and push notifications, with heartbeats and exponential-backoff reconnect that degrades silently to polling. Self-rolled HS256 JWT in HttpOnly cookies with double-submit CSRF, a per-device session registry and token-version revocation, Google OAuth sign-in, and SMTP email OTP. Geospatial artist discovery on native MySQL POINT/SRID 4326 with a SPATIAL-indexed mirror table and a two-stage bounding-box → ST_Distance_Sphere query; Leaflet/OpenStreetMap maps with a hand-built zoom-gated clusterer. Also: two-way Google Calendar sync (libsodium-encrypted refresh tokens), per-user timezone handling, responsive WebP image derivatives with EXIF normalization, structured JSON logging with correlation IDs, an in-app admin monitoring dashboard, and route code-splitting that cut the main bundle 592 → 268 kB (−52% gzip). 341 PHPUnit + 174 Vitest tests, GitHub Actions CI, and scripted backups with a verified restore drill.

- Public Health Information System with Predictive Analytics (Github Repo is Private) — Full-stack health information system built for Rural Health Units and Barangay Health Stations in Balanga City, Bataan. Laravel 12 REST API with Sanctum authentication and policy-based RBAC; React 19 + TypeScript SPA with TanStack Query and Recharts; predictive analytics (z-score outbreak detection, regression-based trend forecasting); hand-built SVG choropleth mapping; multi-format reporting (table/CSV/PDF); ~900 automated tests; CI/CD via GitHub Actions.
- Portfolio Website (michael-cardose.vercel.app) — Built with Next.js, React, and TypeScript; Tailwind CSS and shadcn/ui for styling; Three.js/react-three-fiber for 3D visualizations; dark/light theming; deployed on Vercel.
- GlowSimcha (B2C Web App, Github Repo is Private) —
- AudioScribe (<https://github.com/Michaelcarduce/AudioScribe4.git>) — web app with text-to-speech, speech-to-text, and audio/video processing features, built with the MERN stack, Chakra UI, and deployed on a DigitalOcean server.
- MERN Chat Application (<https://github.com/Michaelcarduce/Chat-App.git>) — real-time chat application using Node.js, Express.js, React.js, MongoDB, and Socket.io for live messaging and notifications.
- University Projects — Library System (Java, MySQL); iTech Payroll System (Visual Basic, Microsoft DBMS); iChurch RFID Attendance & SMS Messaging system (Visual Basic, Microsoft DBMS).